Linear regression: Abalone growth and pH example

Aylin Orozco

2026-05-15

Table of contents

Set up 1

Abalone example 2

- a. Questions and hypotheses . . . 2
- b. Cleaning . . . 2
- c. Exploratory data visualization . . . 2
- d. Abalone model . . . 3
  - Model fitting . . . 3
  - Diagnostics . . . 4
  - Model summary . . . 4
  - Generating predictions . . . 5
  - Why generate model predictions? . . . 6
  - Visualizing model predictions and data . . . 7
  - Creating a table with model coefficients, 95% confidence intervals, and more . . . 8

Set up

```r
library(tidyverse) # general use
library(janitor) # cleaning data frames
library(here) # file/folder organization
library(readxl) # reading .xlsx files
library(ggeffects) # generating model predictions
library(flextable) # creating tables
```

```
# abalone data from Hamilton et al. 2022
abalone <- read_xlsx(here("data", "Abalone IMTA_growth and pH.xlsx"))
```

Abalone example

Data from Hamilton et al. 2022. "Integrated multi-trophic aquaculture mitigates the effects of ocean acidification: Seaweeds raise system pH and improve growth of juvenile abalone." https://doi.org/10.1016/j.aquaculture.2022.738571

a. Questions and hypotheses

Question: How does pH predict abalone growth (measured in change in shell surface area per day, mm^2 d^{-1})?

H_0: **pH has no effect on abalone shell growth**

H_A: **pH has an effect on abalone shell growth**

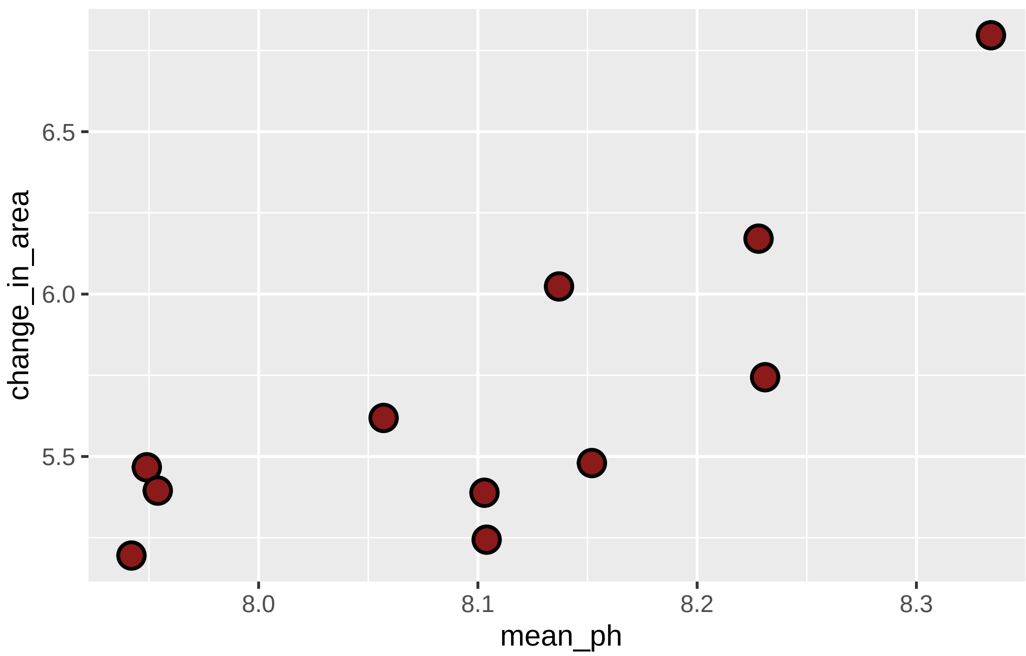
b. Cleaning

```
# creating a new object
abalone_clean <- abalone  |>
  # cleaning names to remove spaces
  clean_names()  |>
  # selecting columns to rename
  select(mean_p_h, change_in_area_mm2_d_1_25) |>
  # changing names of selected columns
  rename(mean_ph = mean_p_h,
         change_in_area = change_in_area_mm2_d_1_25)
```

Don't forget to look at your data! Use `View(abalone_clean)` in the Console.

c. Exploratory data visualization

```
# base layer: ggplot
ggplot(data = abalone_clean, # pulling data from abalone_clean
       aes(x = mean_ph, # x-axis: pH, y-axis: change in shell surface area/day
           y = change_in_area)) +
  # second layer: points to represent observations
  geom_point(size = 4,
             stroke = 1,
             fill = "firebrick4",
             shape = 21)
```



Where does the missing value come from?

insert response here

Don't forget to turn your warnings off either in the individual code chunk or in the YAML at the top of the file.

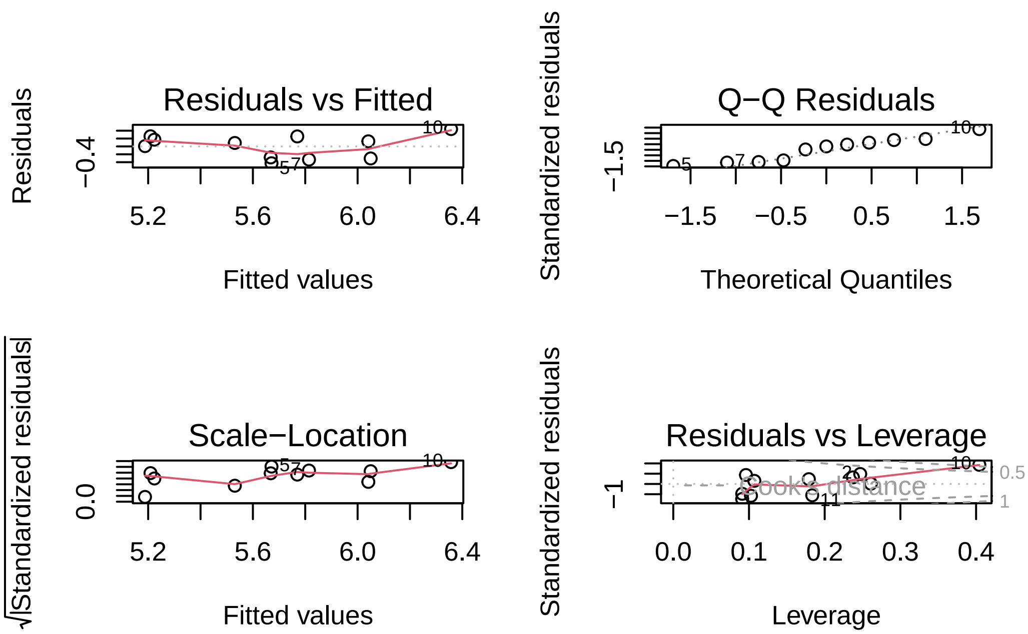
d. Abalone model

Model fitting

```
abalone_model <- lm(
  change_in_area ~ mean_ph, # formula: change in shell SA/day as a function of pH
  data = abalone_clean      # data frame: cleaned abalone data
)
```

Diagnostics

```
par(mfrow = c(2, 2))    # displays plots in a 2x2 grid
plot(abalone_model)     # displays diagnotic plots for model
```



Model summary

```
# more information about the model
summary(abalone_model)
```

```
Call:
```

```
lm(formula = change_in_area ~ mean_ph, data = abalone_clean)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.42714 -0.29282  0.08769  0.21258  0.43965 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)   
(Intercept) -18.5010     6.1601  -3.003  0.01488 * 
mean_ph       2.9827     0.7596   3.926  0.00348 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3063 on 9 degrees of freedom
  (1 observation deleted due to missingness)
Multiple R-squared:  0.6314,	Adjusted R-squared:  0.5905 
F-statistic: 15.42 on 1 and 9 DF,  p-value: 0.003477
```

What is the slope estimate (with standard error?)

β_1: 2.98 $\pm$ 0.76

What is the intercept estimate (with standard error?)

β_0: -18.50 $\pm$ 6.16

Generating predictions

```r
# creating a new object called abalone_preds
abalone_preds <- ggpredict(
  abalone_model,      # model object
  terms = "mean_ph"   # predictor (in quotation marks)
)

# display the predictions
abalone_preds
```

```
# Predicted values of change_in_area

mean_ph | Predicted |     95% CI
--------------------------------
   7.94 |      5.19 | 4.83, 5.54
```

```
   7.95 |      5.21 | 4.86, 5.55
   8.06 |      5.53 | 5.30, 5.76
   8.10 |      5.67 | 5.46, 5.88
   8.10 |      5.67 | 5.46, 5.88
   8.14 |      5.77 | 5.55, 5.98
   8.15 |      5.81 | 5.59, 6.04
   8.33 |      6.36 | 5.92, 6.80
```

Look at the column names using `colnames(abalone_preds)` in the Console.

What are the column names?

`x`, `predicted`, `std.error`, `conf.low`, `conf.high` (not going to worry about `group` right now)

Look at the actual object (by clicking on it in Environment).

Look at the class of the object using `class(abalone_preds)` in the Console.

What is this type of object?

It is a `ggeffects` output and a regular data frame

View `abalone_preds` like a data frame.

Why generate model predictions?

This dataset does not include any actual observations of abalone growth at pH = 8.

However, the model can tell us a prediction for what abalone growth could be if pH = 8 based on the existing data.

Figure out what abalone growth the model would predict at pH = 8.

```
# finding the model prediction at a specific value
ggpredict(
  abalone_model,      # model object
  terms = "mean_ph[8]"   # predictor (in quotation marks) and predictor value in brackets
)
```

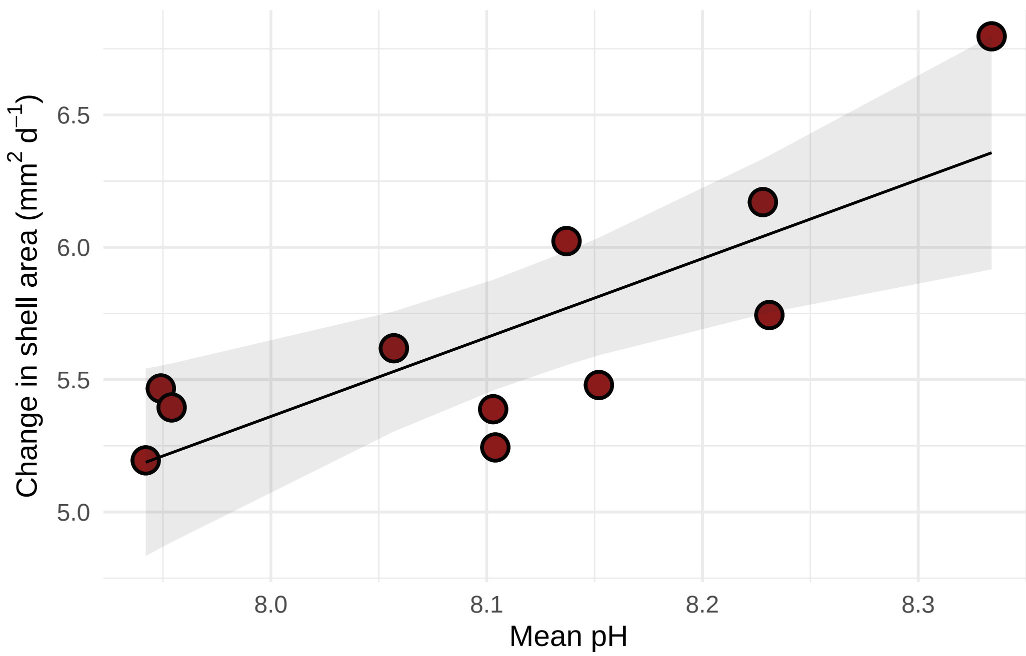
```
# Predicted values of change_in_area

mean_ph | Predicted |     95% CI
--------------------------------
      8 |      5.36 | 5.08, 5.64
```

How would you interpret this?

The model predicts change in abalone shell surface area to be 5.36 [95% CI: 5.08, 5.64], (mm^2 d^{-1})

Visualizing model predictions and data

```r
# base layer: ggplot
# using clean data frame
ggplot(data = abalone_clean,
       aes(x = mean_ph,
           y = change_in_area)) +
  # first layer: points representing abalones
  geom_point(size = 4,
             stroke = 1,
             fill = "firebrick4",
             shape = 21) +
  # second layer: ribbon representing confidence interval
  # using predictions data frame
  geom_ribbon(data = abalone_preds,
              aes(x = x,
                  y = predicted,
                  ymin = conf.low,
                  ymax = conf.high),
              alpha = 0.1) +
  # third layer: line representing model predictions
  # using predictions data frame
  geom_line(data = abalone_preds,
            aes(x = x,
                y = predicted)) +
  # axis labels
  labs(x = "Mean pH",
       y = expression("Change in shell area ("*mm^{2}~d^-1*")")) +
  # theme
  theme_minimal()
```



Creating a table with model coefficients, 95% confidence intervals, and more

Note: this function is from `flextable`.

```
as_flextable(abalone_model)
```

	Estimate	Standard Error	t value	Pr(>|t|)	
(Intercept)	-18.501	6.160	-3.003	0.0149	*
mean_ph	2.983	0.760	3.926	0.0035	**

*Signif. codes: 0 <= '***' < 0.001 < '**' < 0.01 < '*' < 0.05*

Residual standard error: 0.3063 on 9 degrees of freedom

Multiple R-squared: 0.6314, Adjusted R-squared: 0.5905

F-statistic: 15.42 on 9 and 1 DF, p-value: 0.0035

END OF ABALONE EXAMPLE